

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY, CHANGA

B.Tech. (EC/ME/ EE/ CL) SEM - III Examination, 2010

MA - 202 Engineering Mathematics III

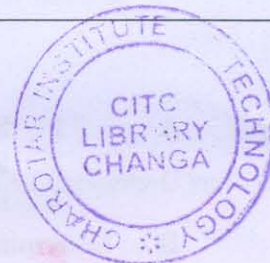
Date: 25.11.10, Thursday

Time: 10:30 a. m to 1:30 p. m.

Maximum Marks:70

Instructions:

1. Attempt all the questions in each section.
2. Assume suitable additional data if required.
3. Figures to the right side indicate the full marks.



Section - I

FOR REFERENCE [03]

Q.1 A State convolution theorem and using it evaluate $L^{-1}\left\{\frac{1}{s^2(s-1)}\right\}$ [03]

B Define Half range Cosine series for the function $f(x)$ on $(0, l)$ and express $f(x) = x$ as a half range Sine series in $0 < x < 2$. [03]

C Obtain Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases}$ [03]

Q.2 A Find the Laplace transforms of the function $f(t) = \begin{cases} 4, & 0 < t < 1 \\ -2, & 1 < t < 3 \\ 5, & t > 3 \end{cases}$ [04]

B Solve $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$ by using Laplace transform. [04]

C Find the Laplace transform of a periodic function of period '2a' define as [04]

$$f(t) = \begin{cases} k, & 0 \leq t < a \\ -k, & a < t < 2a \end{cases}$$

OR

Q.2 A Find the Laplace transforms of the function $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$ [04]

B Solve $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t} \sin t$, $x(0) = 0$, $x'(0) = 1$ by using Laplace transform. [04]

C Find the Laplace transform of $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$ [04]

Q.3 A Solve : $x^3 - 27x + 54 = 0$. [05]

B Solve : $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$. [05]

C Find the Fourier series of the periodic square wave function $f(x)$, and $f(x+2\pi) = f(x)$. [04]

where $f(x) = \begin{cases} -k, & -\pi \leq x < 0 \\ k, & 0 < x < \pi \end{cases}$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

OR

- Q.3 A Solve : $x^3 - 12x + 16 = 0$. [05]
 B Solve: $x^4 + 12x - 5 = 0$. [05]
 C Find the Fourier series to represent $f(x) = 2x - x^2$ in the range $(0, 3)$. [04]

Section - II

- Q.4 A 1 State Green's theorem [03]
 2 State Divergence Theorem
 3 Define Solenoidal Vector Field
 B Determine the constant 'a', so that the vector $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x - az)\hat{k}$ is solenoidal. [03]
 C If $\vec{f} = (xy)\hat{i} + (yz)\hat{j} + (zx)\hat{k}$, then prove $\nabla^2 \vec{f} = 0$. [03]
 Q.5 A Compute $\int_C \vec{f} \cdot d\vec{r}$, where $\vec{f} = \frac{iy - jx}{x^2 + y^2}$ and C is the circle $x^2 + y^2 = 1$ traversed counter clockwise. [04]
 B Evaluate $\oint_C [(xy + y^2)dx + x^2 dy]$ by using Green's theorem where C is closed curve of the region bounded by $y = x^2$ and $y = x$. [04]
 C Using Stoke's theorem, evaluate $\oint_C (x^2 + y^2)dx - 2xydy$, C is taken around the rectangle bounded by $x = \pm a, y = 0, y = b$. [04]

OR

- Q.5 A Use divergence theorem to evaluate $\int_S \vec{A} \cdot d\vec{S}$ over the cube bounded by the planes $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$ where $\vec{A} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$. [04]
 B Evaluate $\int_C [yz dx + (zx + 1)dy + xy dz]$ where C is any path from $(1, 0, 0)$ to $(2, 1, 4)$. [04]
 C Evaluate $\oint_C [(cos x \sin y - xy) dx + \sin x \cos y dy]$ by Green's theorem where C is the circle $x^2 + y^2 = 1$. [04]
 Q.6 A Define Directional Derivative. Find the directional derivative of $f(x, y, z) = x^2 y^2 z^2$ at the point $(1, 1, -1)$ in the direction of tangent to the curve $x = e^t, y = 2 \sin t + 1, z = t - \cos t$ at $t = 0$. [04]
 B A body executes damped forced vibrations given by the equation,
 $\frac{d^2 x}{dt^2} + 8 \frac{dx}{dt} + 64x = 128 \cos 8t$. Solve the equation, when $x = \frac{1}{3}, \frac{dx}{dt} = 0$ at $t = 0$. [05]
 C Show that the frequency of free vibrations in a closed electrical circuit with inductance L and Capacity C in series is $\frac{30}{\pi \sqrt{LC}}$ cycles/minute. [05]

OR

- Q.6**
- A** Define irrotational vector field. Determine the constants a, b, c so that the vector $\vec{F}$ given by $\vec{F} = (2x + 3y + az)\vec{i} + (bx + 2y + 3z)\vec{j} + (2x + cy + 3z)\vec{k}$ is irrotational. [04]
- B** The differential equation of a simple pendulum is $\frac{d^2x}{dt^2} + \omega_0^2 x = F_0 \sin t$. If initially $x = 0, \frac{dx}{dt} = 0$, determining the motion. What happens if $\omega_0 = n$? [05]
- C** An e.m.f. $E \sin pt$ is applied at $t = 0$ to a circuit containing a condenser of capacity C and inductance L in series. The current i satisfies the equation $L \frac{di}{dt} + \frac{1}{C} \int i dt = E \sin pt$. If $p^2 = \frac{1}{LC}$ and initially the charge q and the current i are zero, show that the current at time t is $\frac{Et}{2L} \sin pt$, where $i = \frac{dq}{dt}$. [05]